Sales Forecasting & Demand Intelligence System

Executive Summary — Prepared for the Head of Supply Chain and CFO

July 2026

Executive Summary

Our retail business generated approximately $2.26 million in sales over the past four years across three product categories and four regions. This report summarizes a data-driven analysis covering sales trends, a three-month sales forecast, unusual sales patterns that merit attention, and a grouping of our 17 product lines by demand behavior. Based on these findings, we recommend three specific actions to improve inventory planning and reduce the risk of stockouts and overstock in the coming quarter.

Key Findings

- **Category performance:** Technology is our strongest category at $827,000 (36.6% of sales), ahead of Furniture ($729,000, 32.2%) and Office Supplies ($705,000, 31.2%).
- **Seasonality:** Sales show a clear, recurring pattern — demand spikes in November and December (holiday shopping) and dips sharply in January (post-holiday slowdown), confirmed across multiple years of data.
- **Forecast accuracy:** We tested three forecasting approaches and found a machine learning model to be the most accurate, with a typical forecast error of about 18%, versus roughly 21–22% for the two statistical alternatives.

Three-Month Sales Forecast

Using our most accurate model, we project the following sales for the next three months:

Month	Forecast	Expected Range
Month 1	~$61,700	$50,600 – $72,800
Month 2	~$83,700	$68,700 – $98,800
Month 3	~$83,000	$68,100 – $98,000

In plain terms: we expect a moderate month followed by two stronger months, consistent with the seasonal uptick typically seen in this period. The ranges reflect the model's typical margin of error (about 18%) and should be read as a plausible band, not an exact figure.

Top 3 Anomalies Detected

- **March 22, 2015 —** sales spiked to $37,700, the single highest weekly figure in four years, confirmed independently by two different detection methods. Likely cause: a large one-time bulk or corporate order.
- **November–December 2018 —** three separate weeks (Nov 4, Nov 18, Dec 2) showed unusually high sales, between $29,000 and $36,000. Likely cause: the holiday shopping season, consistent with the seasonal pattern above.
- **Early January (2015 and 2016) —** sales fell to as low as $225–$360 in the first weeks of these years. Likely cause: the typical post-holiday demand slowdown.

Product Demand Segments & Stocking Strategy

Segment	Products	Recommended Strategy

High Volume, Core Growth	Phones, Chairs, Storage, Tables, Binders, Accessories	Maintain strong, consistent stock — these 6 lines drive the majority of revenue.
Explosive Growth	Copiers	Increase order frequency, monitor weekly — demand grew 480% YoY, the fastest of any line.
Steady, Low Volume	Envelopes, Supplies, Furnishings, Labels, Paper, Art, Appliances, Bookcases, Fasteners	Keep lean, order-on-demand — sales are low but predictable.
Declining, High Risk	Machines	Reduce future stock commitments and investigate cause — the only line with falling demand (-30% YoY).

Business Recommendations

1. **Increase inventory and reorder frequency for Copiers.** This line grew 480% year-over-year, the fastest in our portfolio; current ordering may not keep pace with demand, risking lost sales from stockouts.
2. **Review the Machines product line before further large orders.** It is the only line with declining demand (-30% YoY) and the highest sales volatility measured; a closer look at causes (pricing, competition, product lifecycle) is recommended first.
3. **Build additional safety stock ahead of November–December.** Our anomaly analysis flagged unusually high sales in this window across multiple separate years; understocking risks losing sales during the year's peak weeks.

Risk & Limitation

This analysis is based on four years of historical sales data (2015–2018). Our recommended model outperforms traditional forecasting methods on this dataset, but with a relatively short history, this advantage may not hold as strongly for forecasts beyond three months, or if demand patterns shift meaningfully (new products, competitor actions, economic changes). We recommend re-running this analysis quarterly as new data becomes available, and treating any forecast beyond the three-month window with additional caution.